Module 6 Summary

After completing all of the materials, please take a moment to reflect on what was covered. The learning objectives below capture the key takeaways from this module:

- Model bias and model variance to explain the tradeoff between the two.
- List 4 examples of loss functions.
- Define two kinds of model error to evaluate bias versus variance.
- Explain why training error is not good for this purpose.
- Describe the purpose of a training-validation-test split of the dataset.
- Explain the error decomposition into irreducible error, bias, and variance.
- Describe this decomposition with a diagram from the textbook.
- Define AIC and BIC and state their equations.
- Compare and contrast AIC and BIC.
- Identify how best to utilize AIC and BIC.
- State the aim of MDL.
- Study the formula for discussing it as a metric.
- Observe that MDL is equivalent to BIC as a metric.
- State what VC Dimension computes.
- Provide an example with linear regression.
- Define k-fold cross validation.
- Compute an estimate for the expected prediction error using k-fold cross validation.
- Discuss some considerations for picking the number of folds.
- Define Bootstrapping.
- Explain how bootstrapping can construct an estimator for the model.
- Construct the estimator for the expected prediction error.
- Define the 0.632 estimator.

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